Venkata Lakshmi Parimala Pasupuleti

+1-703-261-5410 | lakshmiparimalap@gmail.com | linkedin.com/in/venkata-lakshmi-parimala-pasupuleti | github.com/venkatalakshmiparimala/AgentProbe | Virginia, USA (open to NYC metro)

PROFESSIONAL SUMMARY

AI engineer with 8+ years of production Python engineering and a live open-source LLM evaluation and agent observability toolkit -- AgentProbe (LangGraph, ChromaDB, FastAPI, 5-metric LLM-as-judge eval engine, trace-level agent monitoring, RAG pipeline tracking, OpenTelemetry-compatible tracing semantics). Python (8yr, production) + TypeScript. Shipped 20+ production microservices at enterprise scale (100K+ MAU, 1M+ records) at Entrata across 120+ Agile sprints. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026). Targeting enterprise GenAI deployment and observability roles where I can bring both production engineering depth and first-principles LLM eval expertise.

CORE COMPETENCIES

LLM Observability | Agent Eval Frameworks | LangGraph / LangChain | RAG Pipeline Tracking | OpenTelemetry-aware Tracing | Python (8yr production) | TypeScript | FastAPI | Docker | AWS (S3, EC2, RDS, Glue) | Multi-Agent Orchestration | GenAI Production Deployment | Cross-functional Delivery | Enterprise-Scale Engineering

EXPERIENCE

George Mason University

Aug 2025 -- May 2026

Graduate Teaching Assistant -- Virginia

- Improved student performance in Python and database courses by delivering targeted mentoring across lectures, labs, and office hours for 30+ undergraduates -- stronger grasp of data structures, algorithms, and debugging.
- Accelerated feedback loops by grading assignments with rubric-aligned feedback and developing supplementary Python exercises, reinforcing concepts across sprint-style delivery cycles.

George Mason University

Aug 2024 -- Aug 2025

Research Associate -- Virginia

- Built regression models ($R^2 = 0.977$) in Python and R across WHO and UK Road Safety datasets spanning 190+ countries -- multi-source data collection, transformation, validation, and stakeholder reporting.
- Applied Isolation Forest to automate data integrity checks across 3 research pipelines -- surfaced anomalies that manual review missed, improving downstream model reliability.
- Designed Tableau, ggplot2, and Quarto dashboards for faculty leadership -- improved research comprehension across two cohorts by 40%.

Entrata India (Db Xento Systems)

Oct 2018 -- Aug 2024

Software Engineer -- Pune

- Reduced deployment times by 40% by building 20+ full-stack microservices and modules (Python, PostgreSQL, JavaScript, RabbitMQ) across 120+ Agile sprints for 100K+ monthly active users at enterprise property management companies -- 99.5% uptime sustained over 6+ months through direct production ownership.
- Cut microservices execution time by 50% by leading Docker-based containerization for 30%+ of modules; improved production query performance by 30% by tuning 200+ SQL queries across all client environments.
- Applied Python, PyTorch, and OpenAI API to automate data transformation pipelines across 1.6M financial records -- production AI on regulated data with output validation and quality gates.
- Achieved 100% PHI compliance by migrating all client data to AWS S3 with Libsodium-encrypted pipelines -- zero security incidents; built 5+ internal platform tools cutting resolution time by 40%.
- Eliminated 80% of manual operational workload by designing and automating decision support workflows using Python and PL/SQL.
- Conducted 300+ peer code reviews and mentored 5+ junior engineers, enforcing secure coding practices and TDD/BDD standards across all major releases.

Entrata India (Db Xento Systems)

Jul 2017 -- Sep 2018

Associate Software Engineer -- Pune

- Improved system stability by 35% by developing and optimizing 20+ full-stack property management modules (Python, PostgreSQL) -- 99.5% uptime and 25% faster report generation.
- Reduced critical issue backlog by resolving 100+ high-priority bugs, improving platform reliability for enterprise end users.

PROJECTS

AgentProbe: LLM Evaluation + Agent Observability Toolkit

Live on GitHub | In Progress

Python, LangChain, LangGraph, FastAPI, ChromaDB, Anthropic, OpenAI, Groq, Ollama, React, Docker, GitHub Actions | github.com/venkatalakshmiparimala/AgentProbe

Built a 3-agent LangGraph runtime (Orchestrator, RAG Tracer, Auto Fix) with tool calling, inter-agent state handoffs, and multi-provider LLM routing (Anthropic, OpenAI, Groq, Ollama) -- deployed as a 15-endpoint FastAPI backend running 100% locally at zero cost. Automated agent quality gates with a 5-metric LLM-as-judge eval engine (factual accuracy, hallucination, retrieval, completeness, compliance) and CI/CD integration -- failing runs blocked automatically before production. Delivered end-to-end observability via React dashboard with trace-level agent monitoring -- ChromaDB RAG pipeline tracking per-run metrics, regression drift, retrieval quality, and latency across concurrent agent test suites; structured for integration with external observability backends (OpenTelemetry-compatible tracing semantics).

AI on FHIR: NLP-Driven Medical Records Query System

Flask, LLMs, FHIR, React, Docker, spaCy, BERT, Embeddings, NumPy, Pandas | github.com/venkatalakshmiparimala/AI_on_FHIR

Reduced manual clinical retrieval time by ~60% by building a Flask + React NLP platform translating natural language queries (EN/ES) to FHIR APIs using LLMs, BERT, and embeddings -- HIPAA-compliant RBAC across structured medical records. Demonstrates rapid GenAI prototyping in a regulated domain with compliance constraints designed in from day one.

CrashClarity: Large-Scale Road Accident Analysis (104K Records)

Python, R, scikit-learn, Random Forest, Logistic Regression, ggplot2, Feature Engineering, GitHub Pages

Achieved 85%+ accuracy on 104K+ UK accident records by building a Random Forest + Logistic Regression pipeline with feature engineering across weather, road, and time variables -- deployed interactive dashboard on GitHub Pages for non-technical stakeholders.

EDUCATION

M.S. Data Analytics Engineering — George Mason University, Virginia — 2024 – 2026

GPA: 3.87 | Graduation: May 2026

B.Tech Computer Science & Engineering — Sree Venkateshwara College of Engineering, India — 2013 – 2017

TECHNICAL SKILLS

Languages: Python (8yr production), TypeScript, JavaScript, SQL, R, Bash, PHP, HTML/CSS, C++

AI & Agents: LangChain, LangGraph, Multi-Agent Orchestration, LLM-as-Judge, Eval Frameworks, Agent Observability, Agent Tracing, RAG Pipelines, ChromaDB, Embeddings, LLM Router, Prompt Engineering, OpenAI API, Anthropic API, Groq, Ollama, FastAPI

Cloud & DevOps: AWS (S3, EC2, RDS, Glue), Azure, Docker (production, 30%+ containerization), Kubernetes (orchestration patterns), CI/CD (GitHub Actions), RabbitMQ, Microservices

ML & Data: PyTorch, TensorFlow, XGBoost, scikit-learn, Isolation Forest, Random Forest, Logistic Regression, Feature Engineering, Predictive Modeling, NumPy, Pandas, spaCy, BERT, Hugging Face

Databases: PostgreSQL, MySQL, Oracle, NoSQL -- schema design, query tuning, indexing, audit logging, GraphQL

Testing & Tools: TDD/BDD, Integration Testing, Selenium, Secure Coding, Git, JIRA, Postman, Agile/Scrum